Task 5: Exploratory Data Analysis (EDA)

Titanic Dataset

Elevate Labs Data Analyst Internship

Tools Used: Python, Pandas, Matplotlib, Seaborn

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Objective:
To explore the Titanic dataset using statistical and visual analysis and identify important patterns, relationships, trends, and anomalies affecting passenger survival.

1. Introduction

The Titanic dataset contains information about passengers who travelled on the RMS Titanic. This analysis explores passenger characteristics such as gender, age, passenger class, and fare, along with their relationship to survival.

The objective of this EDA is to identify meaningful patterns and trends in the data using statistical summaries and visualizations.

2. Dataset Overview

The Titanic dataset contains passenger-level information, including demographic details, passenger class, ticket fare, and survival status. These variables provide useful information for understanding the factors associated with passenger survival.

Key Dataset Information

- **Rows:** 891 passengers
- **Columns:** 12
- **Target Variable:** Survived
- **Categorical Variables:** Sex, Embarked, Pclass

Variable	Description
Survived	Survival status (0 = No, 1 = Yes)
Pclass	Passenger class
Sex	Passenger gender
Age	Passenger age
Fare	Ticket fare
SibSp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard

3. Data Quality Analysis

The dataset was examined for missing values, duplicate records, and data types to assess its overall quality before performing further analysis.

Missing Values:

The dataset contains missing values primarily in the Age, Cabin, and Embarked columns. The Cabin column has a particularly high number of missing values.

Duplicate Records:

No significant duplicate records were identified in the dataset.

Data Types:

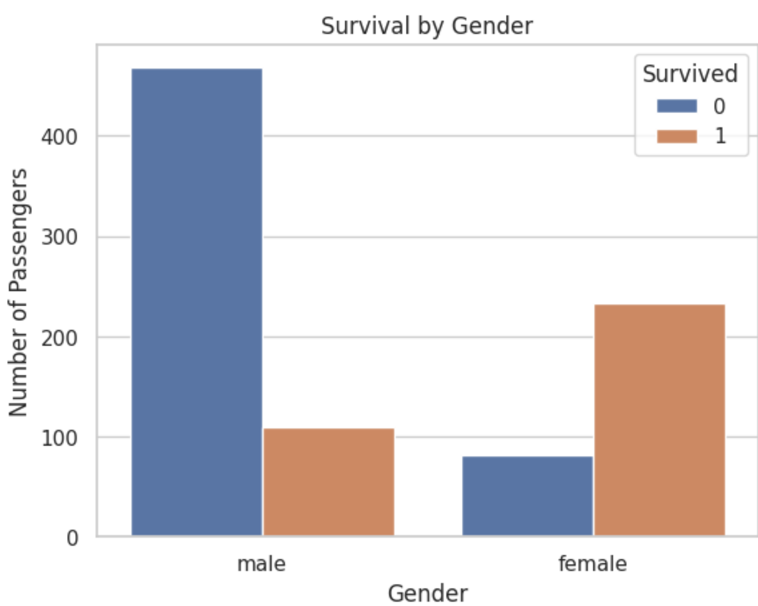
The dataset contains both numerical and categorical variables, making it suitable for statistical analysis and visualization.

Observation:
The missing values, particularly in Age and Cabin, should be considered when interpreting the analysis. However, the dataset remains suitable for exploratory analysis.

4. Survival Analysis:

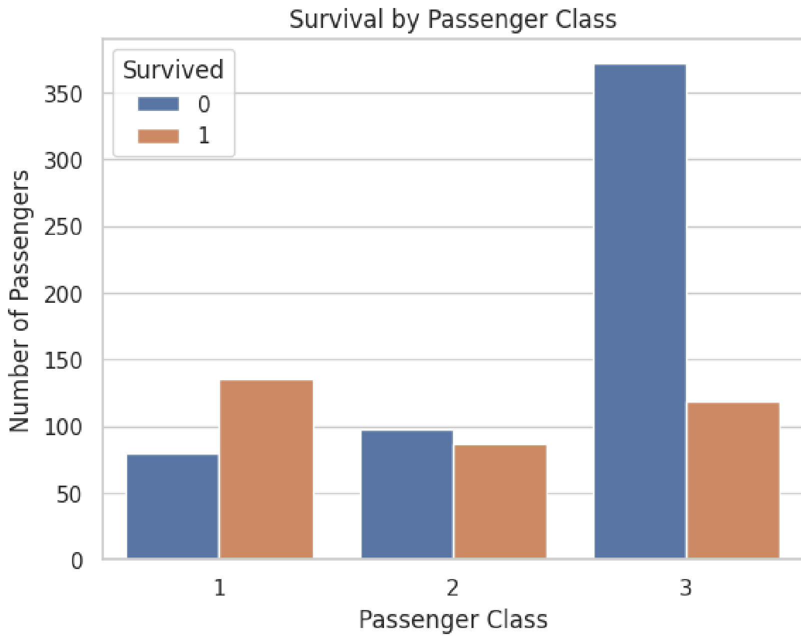
Survival was analyzed across different passenger characteristics to identify factors associatedwith the likelihood of survival.

4.1 Survival by Gender-



Observation:
Female passengers had a significantly higher survival rate than male passengers. This indicates that gender was an important factor associated with survival on the Titanic.

4.2 Survival by Passenger Class-

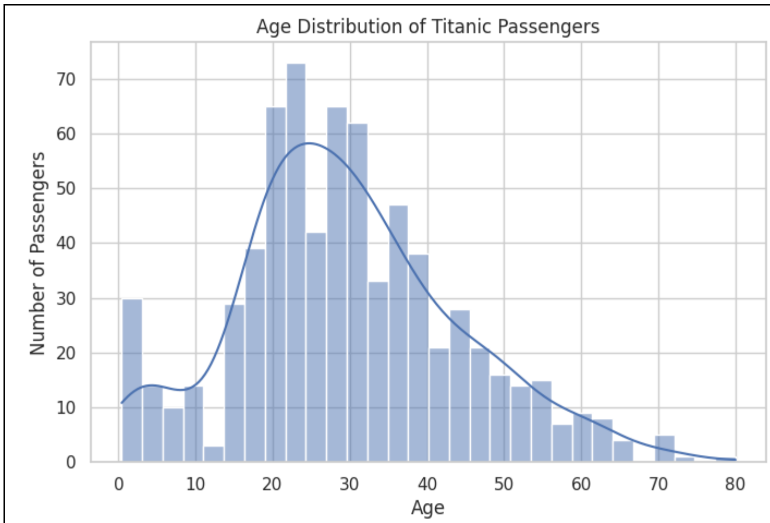


Observation:
First-class passengers had the highest survival rate, while third-class passengers had the lowest survival rate. This suggests that passenger class was strongly associated with survival.

5. Age Analysis

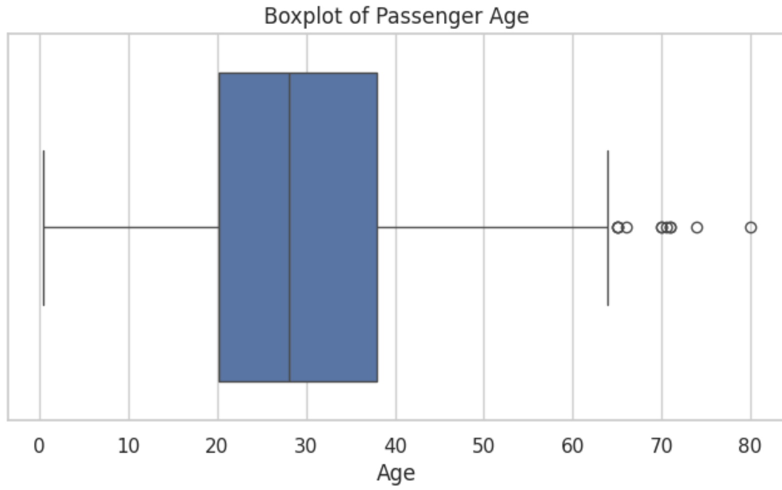
Age distribution was examined using a histogram and boxplot to understand the age range of passengers and identify any unusual values.

Age Distribution:



Observation:
The passengers represented a wide range of ages, with a larger concentration of passengers among children, young adults, and middle-aged individuals.

Age Boxplot-

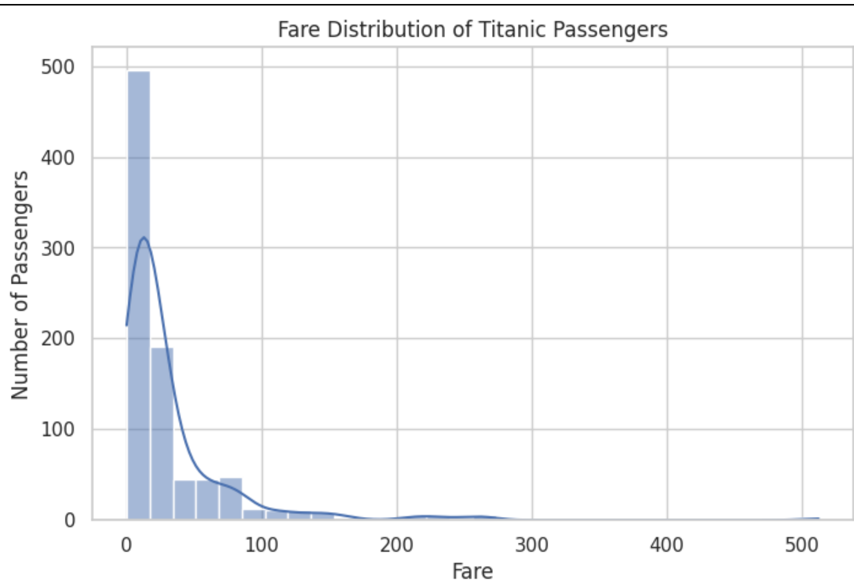


Observation:
The boxplot shows the spread of passenger ages and helps identify potential age outliers. These unusual values were retained because they may represent genuine passengers rather than errors.

6. Fare Analysis

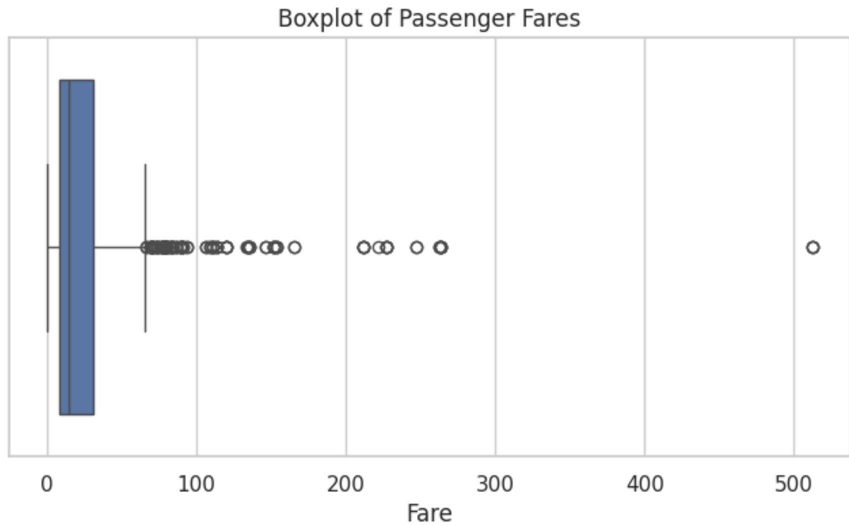
Passenger fares were analyzed using a histogram and boxplot to understand their distribution and identify unusually high or low fare values.

Fare Distribution-



Observation:
Most passengers paid relatively lower fares, while a smaller number of passengers paid substantially higher fares. The distribution is therefore right-skewed.

Fare Boxplot-

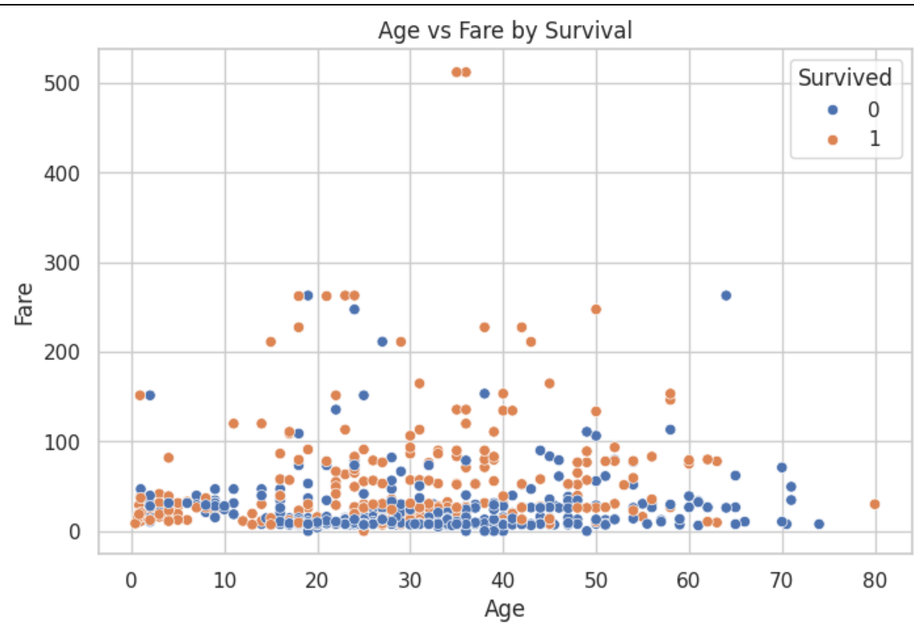


Observation:
The boxplot shows several high-fare values that can be considered potential outliers. These values may be associated with passengers travelling in higher classes or with premium tickets.

7. Relationship Analysis

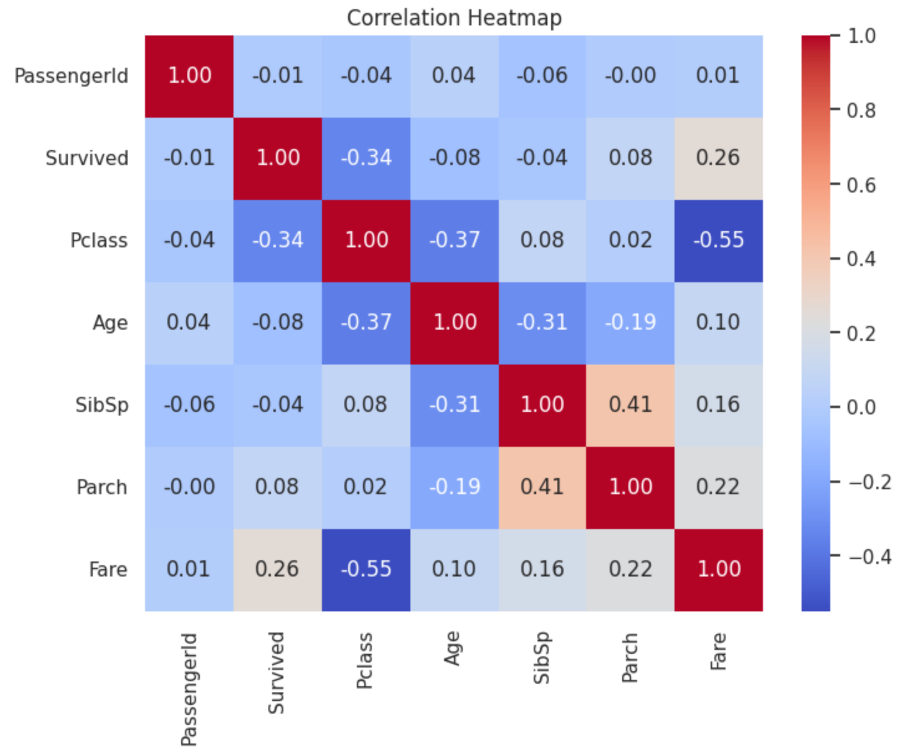
Relationships between numerical variables were explored using scatterplots and a correlation heatmap.

Age vs Fare



Observation:
The scatterplot shows that most passengers paid relatively lower fares across different age groups. There is no strong direct relationship between age and fare, although some passengers paid considerably higher fares.

Correlation Analysis:

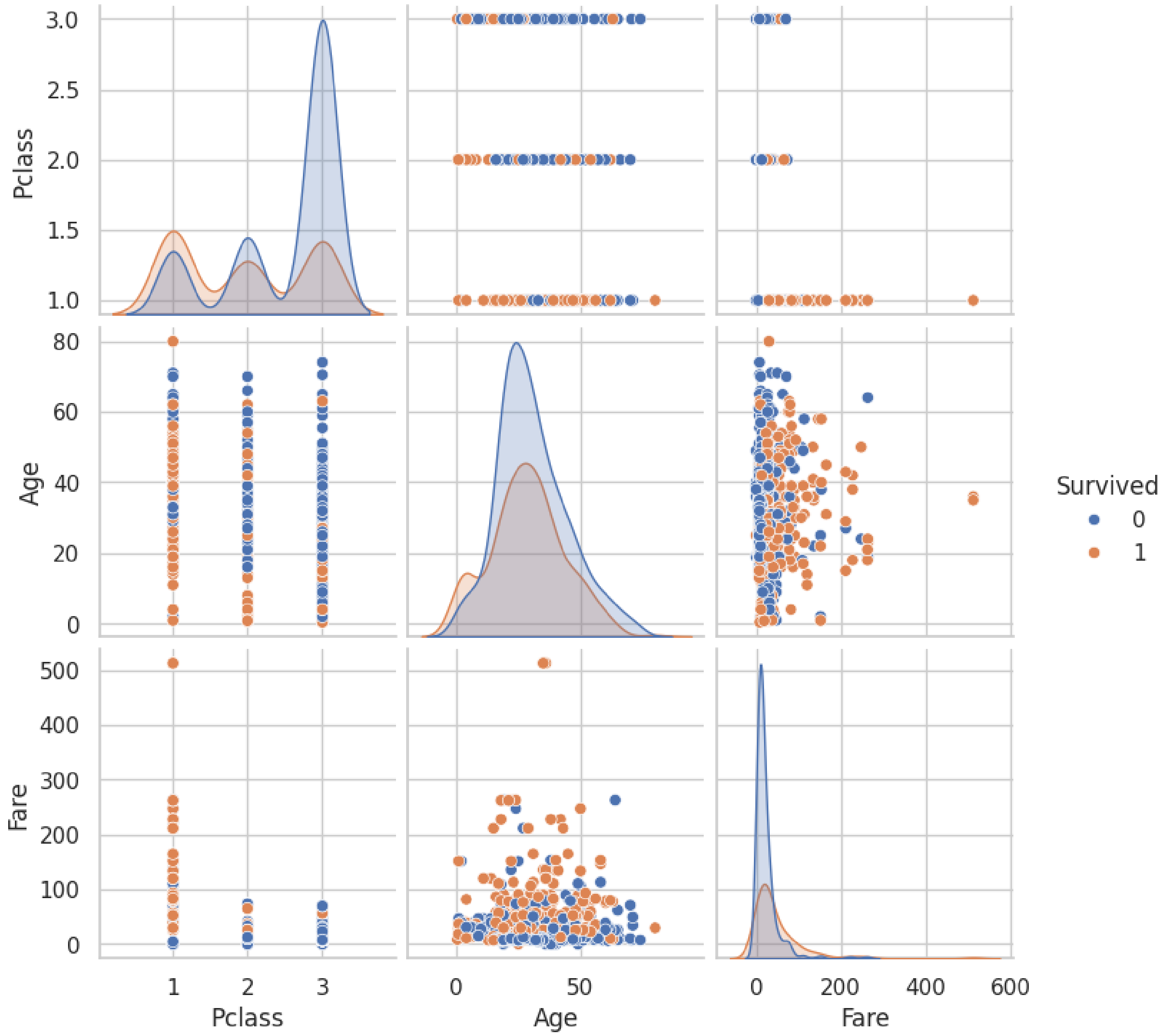


Observation:

The correlation heatmap shows the relationships between numerical variables. Pclass and Fare show a noticeable relationship, while Age has a weaker relationship with most other numerical variables

8. Pairplot Analysis

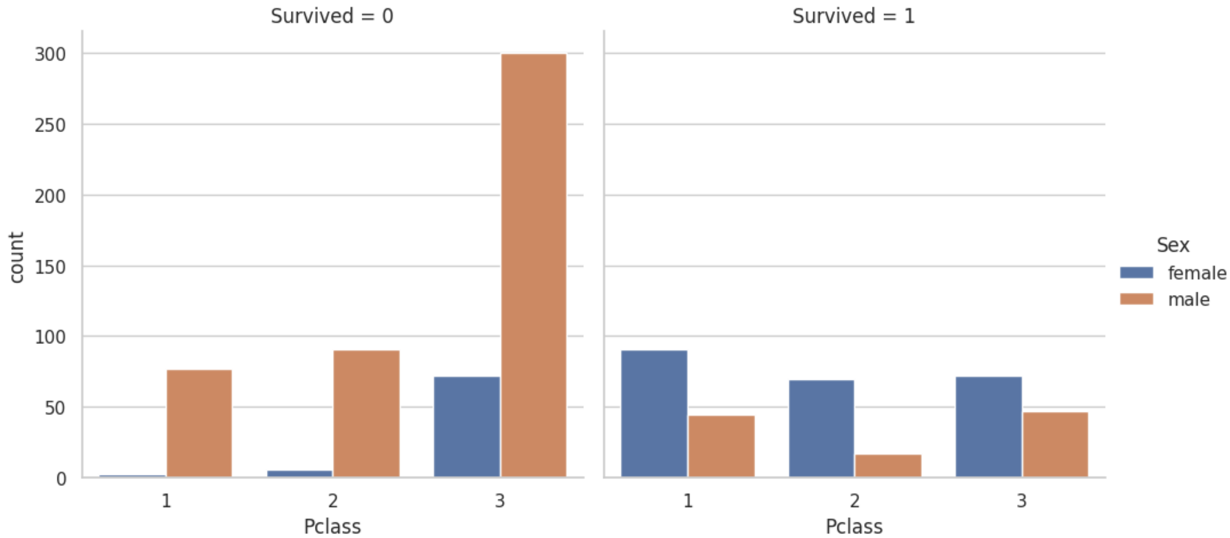
A pairplot was used to examine relationships among Survived, Pclass, Age, and Fare, while using survival status to distinguish the observations.



Observation:
The pairplot provides an overall view of relationships between the selected numerical variables. It also shows that survival patterns vary across passenger class and fare, while age does not show a strong direct relationship with survival.

9. Multivariate Analysis

The combined effect of **gender and passenger class** on survival was analyzed to understand how multiple factors interacted.



Observation:
Survival varied considerably across both gender and passenger class. Female passengers generally had higher survival rates, while passengers in higher classes also showed better survival outcomes.

10. Outlier and Anomaly Analysis

The Interquartile Range (IQR) method was used to identify potential outliers in the Fare and Age variables.

Fare Outliers

Observation:
The fare distribution contains several unusually high values compared with the majority of passengers. These values may represent premium tickets or passengers travelling under different ticket conditions.

Age Outliers

Observation:
A small number of passengers fall outside the typical age range identified by the IQR method. These values appear to be legitimate observations and were therefore not removed.

Overall Observation

Outliers were treated as potentially meaningful observations rather than automatically removing them. This helps preserve the original characteristics of the Titanic dataset.

11. Key Findings and Conclusion

Key Findings

1. **Overall Survival:** A significant proportion of passengers did not survive the Titanic disaster.
2. **Gender:** Female passengers had a considerably higher survival rate than male passengers.
3. **Passenger Class:** First-class passengers had the highest survival rate, while third-class passengers had the lowest.
4. **Age:** Passengers represented a wide range of ages, with most observations concentrated among younger and middle-aged passengers.
5. **Fare:** Most passengers paid lower fares, while a smaller number of passengers had unusually high fares.
6. **Relationships:** Passenger class and fare showed a noticeable relationship, while age and fare did not show a strong direct relationship.
7. **Multivariate Analysis:** Survival varied substantially when gender and passenger class were considered together.

Conclusion

The exploratory analysis of the Titanic dataset reveals that **gender and passenger class were important factors associated with survival.** Female passengers and passengers travelling in higher classes generally had better survival outcomes. The analysis also highlighted differences in fare distribution and identified potential outliers in age and fare.

Overall, the EDA demonstrates how statistical summaries and visualizations can be used to uncover meaningful patterns, relationships, and anomalies in real-world data.